COMP 441 — Software Analysis and Testing

Week 7 — Mutation Testing v1 and Risk-Based Prioritisation

Group A · Track D · TOML Kit

Period	Week 7
Evidence folder	week7
Repository	github.com/the-dezeray/tomlkit
Commit(s)	8c959b5
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Contents	9 sections

This pack collects the week's human-readable report and key evidence. Raw AI-only logs are listed in the inventory and intentionally not printed.

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Source: week7\README.md

Week 7 evidence: Mutation Testing Report v1 (Group A, Track D, tomlkit @ 8c959b5)

Results (AI-only suite `tests_ai/`, mutmut 3.8.0)

	Mutants
Killed	3,087
Timeout (counted as detected)	26
Survived	1,890
No tests (code never executed)	386
Total	**5,389**

Mutation score = (killed + timeout) / total = **57.8%** (62.2% of mutants in executed code). The AI-only suite and manual survivor tests pass together: **404 passed, 3 xfailed**.

Reproduce

```
. .venv/bin/activate && pip install mutmut==3.8.0 hypothesis
mutmut run --max-children 8           # config in pyproject.toml [tool.mutmut]; ~20 min on WSL/Python 3.14
mutmut results --all true > week7/mutmut_results_v1.txt
mutmut show <mutant-name>             # diff of one mutant

# kill tests written for 11 survivors (each fails against its mutant):
cp -r tests_manual mutants/ && cd mutants
MUTANT_UNDER_TEST="tomlkit.items.x‖Array‖insert__mutmut_14" python -m pytest tests_manual -k test_k6_

# risk-based prioritisation + APFD evaluation
pytest -q --cov=tomlkit --cov-context=test --cov-report= tests_ai --durations=0 > week7/durations_ai.txt
pytest --collect-only -q tests_ai | grep :: > week7/collection_order.txt
python week7/scripts/kill_matrix.py mutants week7/mutmut_results_v1.txt 300 week7/kill_matrix.json
python week7/scripts/prioritise.py .coverage week7/radon_cc.json week7/churn_2y.tsv \
       week7/durations_ai.txt week7/kill_matrix.json week7
python week7/scripts/classify_survivors.py <survivor-list> <diff-dir> week7/survivor_categories_v1.csv
```

Survivor root-cause analysis

These eight cases show why the AI-only tests missed representative surviving mutants and how the manual tests target the gaps. The tests are in `tests_manual/test_kill_v1.py`.

Mutant	Why the AI-only tests missed it	Manual test and design technique
`InvalidNumberError.__init__` mutmut 3	Checked the exception type and position, but not the exact, case-sensitive message.	`test_k1_invalid_number_message_and_position_exact` checks the complete message and location (equivalence partitioning).
`Source._to_linecol` mutmut 7	Did not exercise the exact index at a newline boundary.	`test_k2_line_col_at_line_boundary` checks the character before and at the newline (boundary value analysis).
`Parser._parse_escaped_char` mutmut 1	Covered an invalid letter escape, not whitespace after a backslash in a single-line string.	`test_k3_backslash_space_rejected_in_single_line_string` checks rejection, alongside a multiline continuation control (equivalence partitioning).
`parse_rfc3339` mutmut 66	Used timezone offsets below ten hours, missing the two-digit-hour boundary.	`test_k4_two_digit_utc_offsets` checks `+10:00`, `-11:30`, and `+23:59` (boundary value analysis).
`parse_rfc3339` mutmut 163	Checked fractional-second truncation for datetimes, but not the separate time-only parsing branch.	`test_k5_time_fraction_truncated_to_microseconds` checks both time-only parsing paths (equivalence partitioning).
`Array.insert` mutmut 14	Tested positive and negative insertion positions, but not index zero.	`test_k6_insert_at_index_zero` checks the boundary position, resulting values, and round-trip serialization (boundary value analysis).
`Array.__delitem__` mutmut 13	Did not cover stepped slices, a distinct slice behavior from contiguous deletion.	`test_k7_delete_stepped_slice` checks both even- and odd-index deletion (equivalence partitioning).
`Array.add_line` mutmut 8	Tested LF, but not CR or CRLF line breaks.	`test_k8_add_line_comment_rejects_any_line_break` checks all three line-ending forms (equivalence partitioning).

The extra strict xfail in the index-zero insertion tests records a separate spacing defect in the unmutated library; it is not counted as a mutation kill. All 11 targeted manual tests were also run against their named mutants; each mutant was killed.

Test-priority rationale

The risk-based order ranks tests by the additional complexity-weighted, two-year-churn risk they cover; shorter runtime breaks ties. Mutation outcomes are excluded when constructing the order, keeping the APFD comparison out-of-sample. The top-ranked parse test covers the most new risk, especially `Container.append` (complexity 74); rank 2 targets `Container._replace_at` (41), and rank 4 exercises `InlineTable.as_string` (30). This puts high-risk container and formatting behavior early without tuning the ranking to the sampled mutants.

The 300-mutant kill-matrix sample had one mutant that did not remain killed in the fresh v1 run; it was excluded. Of the remaining sample, 258 mutants had matching test identifiers for APFD. The risk-based order achieved APFD **0.7722**, compared with **0.6369** for collection order and **0.6911** for the mean of 200 shuffled orders. It detected half of those mutants within 16 tests, versus 51 in collection order. The full ranking and evaluation are in `priority_order.csv` and `prioritisation_eval.json`.

Files

- `mutmut_results_v1.txt`, `mutation_v1_summary.json` - every mutant's status; score per module
- `survivor_categories_v1.csv` - all 1,890 survivors with diff and heuristic category
- `kill_matrix.json` - which AI tests fail for a seeded sample of 300 mutants; one stale mutant is excluded from APFD
- `priority_order.csv` - ranked execution order for the suite
- `prioritisation_eval.json`, `detection_curve.csv` - APFD of risk-based vs pytest order vs random
- `churn_2y.tsv` - lines changed per file since 2024-10-18 (git log --numstat)

Mutation v1 summary by module

Source: week7\mutation_v1_summary.json

Module	Total	Killed	Timeout	Survived	No tests	Score	Covered
TOTAL	5,389	3,087	26	1,890	386	57.8%	62.2%
api	149	118	7	24	0	83.9%	83.9%
container	1,261	595	0	597	69	47.2%	49.9%
exceptions	174	58	0	94	22	33.3%	38.2%
items	2,000	1,082	1	653	264	54.1%	62.4%
parser	1,387	876	11	469	31	64.0%	65.4%
source	127	103	2	22	0	82.7%	82.7%
toml_file	73	53	0	20	0	72.6%	72.6%
_utils	218	202	5	11	0	95.0%	95.0%

Detection curve (risk-based order)

Source: week7\detection_curve.csv

order	tests_run	pct_mutants_detected
Risk-based	1	14.7
Risk-based	5	25.6
Risk-based	10	43.0
Risk-based	20	54.7
Risk-based	30	65.9
Risk-based	40	70.5
Risk-based	50	72.1
Risk-based	75	82.2
Risk-based	100	89.5
Risk-based	150	100.0
Risk-based	200	100.0
Risk-based	250	100.0
Risk-based	300	100.0
Risk-based	149	100.0
Default order	1	14.7
Default order	5	17.8

order	tests_run	pct_mutants_detected
Default order	10	27.5
Default order	20	31.0
Default order	30	40.7
Default order	40	45.3
Default order	50	49.2
Default order	75	67.1
Default order	100	82.9
Default order	150	100.0
Default order	200	100.0
Default order	250	100.0
Default order	300	100.0
Default order	149	100.0

Prioritisation evaluation (APFD)

Source: week7\prioritisation_eval.json

Key	Value
tests	149
mutants_in_matrix	258
apfd_risk_based	0.7722
apfd_default_order	0.6369
apfd_random_mean	0.6911
apfd_random_min	0.6272
apfd_random_max	0.7532
tests_to_detect_50pct_risk_based	16
tests_to_detect_90pct_risk_based	106
tests_to_detect_50pct_default_order	51
tests_to_detect_90pct_default_order	120

Risk-based test order (first 20 tests)

Source: week7\priority_order.csv

rank	test	new_risk_covered	functions_executed	top_function
1	tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_returns_document	2417.2	47	Container.append (CC 74)
2	tests_ai/test_ai_container.py::TestEditingPreservesStyle::test_update_value_keeps_comment	552.4	5	Container._replace_at (CC 41)
3	tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[[[aot]]\nx = 1\n\n[[aot]]\nx = 2\n]	552.1	14	Container.append (CC 74)
4	tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[inline = { a = 1, b = "two" }\n]	377.1	8	InlineTable.as_string (CC 30)
5	tests_ai/test_ai_items.py::TestArray::test_delitem_last	246.6	1	Array.__delitem__ (CC 36)
6	tests_ai/test_ai_items.py::TestArray::test_add_line_with_comment	239.8	3	Array.add_line (CC 24)
7	tests_ai/test_ai_container.py::TestDottedAndOutOfOrder::test_out_of_order_add_key	203.1	4	Container.append (CC 74)
8	tests_ai/test_ai_api.py::TestEncoders::test_register_and_unregister	187.5	4	item (CC 31)
9	tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_identical	180.1	8	Container._render_table (CC 31)
10	tests_ai/test_ai_container.py::TestEditingPreservesStyle::test_add_scalar_goes_before_tables	176.9	3	Container.append (CC 74)
11	tests_ai/test_ai_api.py::TestParseAndLoads::test_loads_is_alias_of_parse	144.1	5	_equal_with_nan (CC 13)
12	tests_ai/test_ai_container.py::TestDottedAndOutOfOrder::test_dotted_keys_access	137.6	9	Container.append (CC 74)
13	tests_ai/test_ai_parser.py::TestStrings::test_string_values[a = "\\u00e9\\U0001F600"-\xe9\U0001f600]	135.9	4	Parser._parse_escaped_char (CC 12)
14	tests_ai/test_ai_items.py::TestIntegerFloat::test_integer_arithmetic_returns_integer	102.8	7	Integer._new (CC 3)

rank	test	new_risk_covered	functions_executed	top_function
15	tests_ai/test_ai_container.py::TestDottedAndOutOfOrder::test_out_of_order_delete	98.3	3	OutOfOrderTableProxy.__delitem__ (CC 7)
16	tests_ai/test_ai_items.py::TestIntegerFloat::test_integer_bitwise	68.5	5	Integer.__rshift__ (CC 2)
17	tests_ai/test_ai_api.py::TestKeyFactory::test_keys_in_document	67.6	5	Container.append (CC 74)
18	tests_ai/test_ai_items.py::TestIntegerFloat::test_float_arithmetic	54.8	4	Float.__mul__ (CC 2)
19	tests_ai/test_ai_api.py::TestKeyFactory::test_dotted_key	53.0	2	DottedKey.__init__ (CC 5)
20	tests_ai/test_ai_container.py::TestDottedAndOutOfOrder::test_out_of_order_table_proxy	52.7	6	Container.append (CC 74)

Showing the first 20 of 149 rows. Full file: `week7\priority_order.csv`.

Survivor categories

Source: week7\survivor_categories_v1.csv

Value	Count	Share
Argument dropped or set to None	756	40.0%
Boolean logic (and/or, True/False, not)	271	14.3%
Exception message or construction	256	13.5%
String literal changed (XX..XX / case)	196	10.4%
Arithmetic / assignment operator	147	7.8%
Boundary / off-by-one (< vs <=, +1, index)	122	6.5%
Control flow (continue/break/return)	96	5.1%
Other	46	2.4%
Total	**1890**	**100.0%**

1890 rows summarised by `category`.

Pytest collection order (first 20 tests)

Source: week7\collection_order.txt

```
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_returns_document
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_values
tests_ai/test_ai_api.py::TestParseAndLoads::test_loads_is_alias_of_parse
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_accepts_bytes
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_empty_string
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_invalid_raises
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_identical
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[a = 1\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[a = 1 # comment\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[[t]\n  x = 'y'\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[arr = [\n  1,\n  2,\n]\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[inline = { a = 1, b = "two" }\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[[[aot]]\nx = 1\n\n[[aot]]\nx = 2\n]
tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[\n\n# only comments\n\n]
tests_ai/test_ai_api.py::TestDumps::test_dumps_plain_dict
tests_ai/test_ai_api.py::TestDumps::test_dumps_nested_dict_creates_table
tests_ai/test_ai_api.py::TestDumps::test_dumps_sort_keys
tests_ai/test_ai_api.py::TestDumps::test_dumps_sort_keys_on_document
tests_ai/test_ai_api.py::TestDumps::test_dumps_rejects_non_mapping
tests_ai/test_ai_api.py::TestDumps::test_dumps_table_item
```

Showing the first 20 of 388 lines. Full file: `week7\collection_order.txt`.

AI test durations (first 20 lines)

Source: week7\durations_ai.txt

```
.....................................................................  [ 18%]
.....................................................................  [ 37%]
.....................................................................  [ 55%]
x....................................................................  [ 74%]
....................x................................................  [ 92%]
........................                                               [100%]
============================== warnings summary ==============================
tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_returns_document
  /mnt/c/Users/macha/Documents/comp_441/tomlkit/.venv/lib/python3.14/site-packages/coverage/control.py:803: CoverageWarn
  ing: Dynamic contexts aren't supported with core=sysmon; context data may be incomplete (no-sysmon-context); see https:/
  /coverage.readthedocs.io/en/7.16.2/messages.html#warning-no-sysmon-context
    self._warn(
```

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================== slowest durations ===============================
0.14s setup    tests_ai/test_ai_api.py::TestParseAndLoads::test_parse_returns_document
0.03s call     tests_ai/test_ai_items.py::TestPickle::test_bool_equals_bool
0.03s teardown tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[[t]\n  x = 'y'\n]
0.02s teardown tests_ai/test_ai_items.py::TestArray::test_clear
0.02s teardown tests_ai/test_ai_container.py::TestEditingPreservesStyle::test_update_value_keeps_comment
0.02s teardown tests_ai/test_ai_api.py::TestRoundTrip::test_round_trip_preserves_formatting[a = 1\n]
0.02s teardown tests_ai/test_ai_parser.py::TestInlineTables::test_duplicate_key_in_inline
```

Showing the first 20 of 194 lines. Full file: `week7\durations_ai.txt`.

The full 5,389-line mutmut output is `week7/mutmut_results_v1.txt`; the 300-mutant kill matrix is `week7/kill_matrix.json`. The original Week 7 PDF report is `week7/mutation-testing-report-v1.pdf`; it is kept as a raw artifact and is not appended here because it duplicates this README and uses legacy font encoding.

Raw evidence inventory

The pack prints the readable report and key summaries. Large raw artefacts are kept in the repository and listed here so a marker can verify them without paging through thousands of lines. Items tagged **[AI-only]** are deliberately not printed: they are the AI baseline's raw output, not the final hybrid result.

File	Size	Tag
week7\churn_2y.tsv	279 B	
week7\collection_order.txt	26 KB	
week7\detection_curve.csv	658 B	
week7\durations_ai.txt	16 KB	
week7\kill_matrix.json	325 KB	
week7\mutation-testing-report-v1.pdf	126 KB	[v1 baseline]
week7\mutation_v1_summary.json	1 KB	[v1 baseline]
week7\mutmut_results_v1.txt	332 KB	[v1 baseline]
week7\prioritisation_eval.json	383 B	
week7\priority_order.csv	15 KB	
week7\radon_cc.json	146 KB	
week7\README.md	6 KB	
week7\scripts\classify_survivors.py	2 KB	
week7\scripts\kill_matrix.py	1 KB	
week7\scripts\prioritise.py	5 KB	
week7\survivor_categories_v1.csv	314 KB	[v1 baseline]